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ALL INDIA TEST SERIES

FIITJEE

JEE(Main)-2016

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II
(Main)

S. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	D	B	B
2.	A	C	A
3.	C	B	C
4.	D	A	B
5.	B	D	A
6.	C	–	D
7.	C	B	C
8.	B	A	A
9.	A	A	C
10.	B	D	B
11.	A	B	C
12.	A	C	A
13.	B	B	D
14.	C	C	B
15.	C	D	C
16.	D	A	C
17.	A	C	D
18.	A	B	A
19.	D	A	B
20.	B	B	A
21.	C	A	A
22.	C	C	D
23.	A	A	B
24.	A	B	A
25.	D	A	B
26.	C	B	C
27.	D	B	C
28.	B	B	B
29.	B	A	A
30.	A	C	D

Physics

PART – I

SECTION - A

1. D

$\vec{F} = I\left(\vec{l} \times \vec{B}\right)$

Here $\vec{l}$ is directed from end A to end B of the wire.

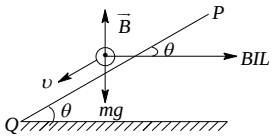
Here $\left|\vec{l}\right| = \frac{L}{2} and\ \theta = 90°.....F = I \times \frac{L}{2} \times B = \frac{BIL}{2}$

2. A

Force on a current carrying conductor is $\vec{F} = i\left(\vec{r} \times \vec{B}\right)$

3. C

$BIL\cos\theta = mg\sin\theta\ or B = \frac{mg}{IL}\frac{\sin\theta}{\cos\theta} = \frac{mg}{IL}\tan\theta$



4. D

The magnetic forces acing on the three sides of the triangle are as follows:

$F_{CD} = 0\ as\ CD$ is parallel to $\vec{B}$

$F_{AC} = ilB\sin 60° = \frac{\sqrt{3}}{2}ilB$ (Perpendicular to the plane of diagram and outwards) as the side AC makes an angle $60°$ with $\vec{B}$.

$F_{AD} = ilB\sin 120° = \frac{\sqrt{3}}{2}ilB$ (Perpendicular to the plane of diagram and outwards) as the side AD makes an angle $120°$ with $\vec{B}$.

Therefore, net force on the loop is

$F = 0 + \frac{\sqrt{3}}{2}ilB + \frac{\sqrt{3}}{2}ilB = \sqrt{3}\,ilB$ perpendicular to the plane of the loop, outwards.

5. B

The magnetic moment of straight wire, $m = \left(q_m\right)\left(l\right)$

The magnetic moment of the semicircular wire, $m' = \left(q_m\right)\left(2r\right)$

$\therefore \frac{m'}{m} = \frac{2r}{l} = \frac{2r}{\pi r} = \frac{2}{\pi} \therefore m' = \frac{2m}{\pi}$

6. C

$T = 2\pi\sqrt{\frac{I}{mB_H}}; T' = 2\pi\sqrt{\frac{2I}{mB_H}} = \sqrt{2}T$

7. C

For solenoid, $B = \mu_0 ni = \mu_0 \frac{N}{l} i$

$\phi = NBA = N\left[\mu_0 \frac{N}{l} i\right] A = \mu_0 \frac{N^2 A}{l} i$

Also $\phi = Li \therefore L = \dfrac{\mu_0 N^2 A}{l}$

So $L = 4\pi \times 10^{-7} \times \dfrac{(1000)^2}{1} \times 10 \times 10^{-4}$

$= 0.0012566H = 1.26mH$

8. B

$X_L = \dfrac{1}{\omega C} = \dfrac{1}{2\pi \nu C} = \dfrac{1}{2\pi \times 50 \times 10 \times 10^{-6}} = 318.31\Omega$

$Z = \sqrt{R^2 + X^2} = \sqrt{R^2 + (X_L - X_C)^2}$

$= \sqrt{0^2 + (100\pi - 318.31)^2} = 4.16\Omega$

9. A

$i = 45° \sin r = \dfrac{\sin i}{n} = \dfrac{\sin 45°}{2} = \dfrac{1}{2\sqrt{2}}$

Lateral shift $= \dfrac{t}{\cos r} \sin(i - r) = t \sin i - t \cos i \ \tan r$

$= \sqrt{2}\,\dfrac{1}{\sqrt{2}} - \dfrac{1}{\sqrt{7}} = \dfrac{\sqrt{7} - 1}{\sqrt{7}} \approx 0.62\,cm$

10. B

The relative velocity of the diving bird relative to the fish $= 4 + \upsilon_a \times \dfrac{4}{3} = 16\,cm\ s^{-1}$

$\therefore \upsilon_a \times \dfrac{4}{3} = 12 \qquad \therefore \upsilon_B = \dfrac{36}{4} = 9\ cm\ s^{-1}$

11. A

Surface AB forms the image of O at v_1 from it. We have $\dfrac{1.5}{v_1} - \dfrac{1}{-4R} = \dfrac{1.5 - 1}{R}$

$\therefore$ For the surface CD, μ_2 = object distance= separation between the two surfaces $-v_1 = -3R + 6R = +3R$

Hence v_2 the image distance due to surface CD is got by

Giving $\dfrac{1}{v_2} = \dfrac{1}{4R} + \dfrac{1}{2R} = \dfrac{3}{4R}$

$or\ v_2 = \dfrac{4R}{3}$

A real image is formed at a distance $\dfrac{4R}{3}$ from the surface CD to its right

12. A

$\mu_1 = -40\,cm \quad f_1 = 30\,cm$

$\therefore \quad v_1 = 120\,cm$ From $v = \dfrac{u\,f}{u + f}$

u_2 For the II lens= $(-d + 120)\,cm$ and $f_2 = 50\,cm$

$\therefore\ v_2 = \dfrac{-50(d-120)}{-(d-170)} = \dfrac{100}{3}$

By solving, we get $d = 20\,cm$

13. B

$I = I_{COM} + mr^2 \Rightarrow 3(10)^2 = m(k)^2 + m(6)^2$

$\therefore\ k = 8\,cm$

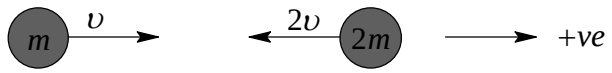
14. C

$I = I_{COM} + mr^2 \Rightarrow \dfrac{7}{12}ml^2 = \dfrac{ml^2}{12} + mr^2$

$\therefore\ r^2 = \dfrac{l^2}{2} \Rightarrow r = \dfrac{l}{\sqrt{2}}$

15. C

$u_1 = +\upsilon$ $\quad u_2 = -2\upsilon$ $\quad m_1 = m$ $\quad m_2 = 2m$



$\upsilon_1 = \left(\dfrac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \left(\dfrac{2m_2}{m_1 + m_2}\right)u_2$

and $\upsilon_2 = \left(\dfrac{m_2 - m_1}{m_1 + m_2}\right)u_2 + \left(\dfrac{2m_1}{m_1 + m_2}\right)u_1$

$\therefore \upsilon_2 = \left(\dfrac{2m - m}{2m + m}\right)(-2\upsilon) + \left(\dfrac{2m}{2m + m}\right)(\upsilon)$

$= -\dfrac{2}{3}\upsilon + \dfrac{2}{3}\upsilon = 0$

16. D

When the extension is maximum, their velocities are equal. From the law of conservation of Momentum,

$P_f = p_i \Rightarrow (6)\upsilon + (3)\upsilon = 6(2) + 3(-1)$

$\upsilon = 1\,m\,s^{-1}$

This energy is also conserved

$E_f = E_1 \Rightarrow \dfrac{1}{2}(6)(1)^2 + \dfrac{1}{2}(3)(1)^2 + \dfrac{1}{2}Kx_m^2$

$= \dfrac{1}{2}(6)(2)^2 + \dfrac{1}{2}(3)(1)^2$

$3 + 1.5 + \dfrac{1}{2}(200)x_m^2 = 12 + 1.5$

$100x_m^2 = 9 \Rightarrow x_m^2 = 0.09 \Rightarrow x_m = 0.3m = 30\,cm$

17. A

$mgh = \dfrac{1}{2}mu_{\min}^2$

$u_{\min} = \sqrt{5gr}$

$\therefore h = 2.5\,r$

18. A

$$|W_f| = \frac{1}{2}m\upsilon^2 - 0 = \frac{1}{2}m\upsilon^2 = \frac{p^2}{2m}$$

$$\therefore \mu mgx = \frac{p^2}{2m} \Rightarrow \mu = \frac{p^2}{2gm^2x}$$

19. D

Work done on both the blocks is same and total work done by spring is $-\frac{1}{2}kx^2$

Hence work done on each blocks is $-\frac{1}{4}kx^2$

20. B

We know that $F = ma$

$$\therefore 200 = m_1 \times 80 \Rightarrow m_1 = \frac{200}{80} = 2.5\,kg$$

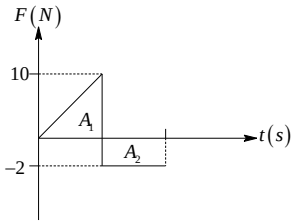
and $200 = m_2 \times 240 \Rightarrow m_2 = \frac{200}{240} = 0.833\,kg$

When joined together, $m = m_1 + m_2 = \frac{20}{6}kg$

Final acceleration is

$$a = \frac{F}{m} = \frac{200}{20/6} = 60\,ms^{-2}$$

21. C



Change in momentum= impulse
= Net area under $F-t$ graph

$$= \left(\frac{1}{2}\times 10\times 10\right) - (10\times 2) = 30\,kg\ ms^{-1}$$

22. C

Let density of material of sphere (in $g\,cm^{-3}$) be p Applying the condition of floatation,

Weight = Upthrust $\Rightarrow pVg = \frac{V}{2}p_{oil}g + \frac{V}{2}p_{Hg}g$

$$p = \frac{p_{oil}}{2} + \frac{p_{Hg}}{2} = \frac{0.8}{2} + \frac{13.6}{2} = 7.2\,g\,cm^{-3}$$

23. A

$$\frac{Volume\ submerged}{Volume} = \frac{p_s}{p_L} = \frac{1}{2}$$

The result is independent of the value of g_{eff}

24. A
This is the effective acceleration due to gravity
$$T' = 2\pi\sqrt{\frac{1}{g_{eff}}} = 2\pi\sqrt{\frac{pl}{(p-\sigma)g}} = \left(\frac{p}{p-\sigma}\right)^{1/2} T$$
$$\left(\because T = 2\pi\sqrt{\frac{l}{g}}\right)$$

25. D
$C_p - C_v = R$ for all gases

26. C
$W_{12} = p(V_2 - V_1)$ ($\because$ constant pressure process)
$= nR(T_2 - T_1) = 2R(500-300) = 400R$

27. D
The force of buoyancy arises due to pressure difference.

28. B
Velocity of knife edge is $\upsilon = \sqrt{2gh}$
Retardation, $a = \frac{\upsilon^2}{2^5} = \frac{gh}{5}$
Average resistance $R - mg = ma$
$$R = mg + m\left(\frac{gh}{s}\right)$$
$$= mg\left(1+\frac{h}{s}\right)$$

29. B
$F - mg\sin\theta = mA$ ..... (1)
$$F - \left(\frac{m}{L}x\,g\sin\theta\right) - T\frac{m}{L}x\left(\frac{F - mg\sin\theta}{m}\right)$$ ..... (2)
Buy solving equation $(1)\&(2)$
$$T = f\left(1-\frac{x}{L}\right)$$

30. A
Work done on bringing a test charge from (1) to (2) is
Is $V_1 - V_2 = \frac{q_1}{4\pi \in_0}\left(\frac{1}{a}-\frac{1}{b}\right)$

Chemistry

PART – II

SECTION - A

1. B

Moles of $NO_2 = \frac{112}{22400} moles = \frac{1}{200} moles$

$wt\ of\ NO_2 = \frac{1}{200} \times 46g$

Volume of liquid $NO_2 = \frac{46}{200} / 1.15 = 0.2\ mL$

2. C

$HCOOH \longrightarrow CO(g) + H_2O(l)$

$a\ mole \qquad 0$

$0 \qquad a\ mole$

$H_2C_2O_4 \longrightarrow CO(g) + CO_2(g) + H_2O(l)$

$b\ mole \qquad 0 \qquad 0$

$0 \qquad b\ mole \qquad b\ mole$

$CO + CO + CO_2 = (a + 2b)$

$\frac{1}{6} \times (a + 2b) = b$

$a + 2b = 6b$

$a / b = 4$

4. A

$A(g) \longrightarrow 2B(g)$

$a\ molL^{-1} \qquad 0 \ \ldots\ldots\ldots\ldots\ldots\ldots\ t = 0$

$(a - x)\ mol\ H^{-1} \qquad 2x\ mol\ L^{-1} \ \ldots\ldots\ t = t$

$\frac{a - x}{2x} = \frac{3}{1} \Rightarrow 2a - 2x = 6x$

$8x = 2a \Rightarrow x = a / 4$

$t = \frac{2.303}{2.303 \times 10^{-2}} \log \frac{a}{a - \frac{a}{4}}$

$t = 10^2 \log \frac{4}{3}$

$12.5\ \text{min}$

5. D

$A(g) \rightleftharpoons 2B(g)$

$at\ P_1 \qquad 1 - \frac{1}{3} \qquad \frac{2}{3} \ldots\ldots\ldots\ldots K_p = \frac{P_1}{2}$

$at\ P_2 \qquad 1 - \frac{2}{3} \qquad \frac{4}{3} \ldots\ldots\ldots\ldots K_p = \frac{16 \times P_2}{5}$

$$\frac{P_1}{2}=\frac{16\times P_2}{5}\ \Rightarrow \frac{P_1}{P_2}=\frac{32}{5}$$

6.

$$\left(H^+\right)_{initially}=10^{-2}$$

$$\left(H^+\right)_{finally}=\frac{10^{-2}}{2}$$

$$pH=-\log\frac{10^{-2}}{2}=2.3\left(\text{approximatly}\right)$$

7. B

fcc unit cell is formed.

$$\frac{\sqrt{3}a}{4}=2R=\text{distance b/w corner and T.V}$$

$$\text{or}\ \sqrt{3}\,a=8R$$

8. A

1 m sucrose solution means 1 mole sucrose $\left(C_{12}H_{22}O_{11}\right)/1000\ g\ H_2O$

$342\ g\ sucrose/1342\ g\ solution$

$254.8\ g$ sucrose $/1000\ g$ solution $/745.2\ g\ H_2O$

$$3.534=\frac{1000\times1.86\times254.8}{342\times W}\ \Rightarrow W=392\ g$$

wt of ice separated out $=\left(745-392\right)g\ \Rightarrow\ 353g$

9. A

$$H_2O\left(l\right)\longrightarrow 2H_2\left(g\right)+O_2\left(g\right)$$

$$\frac{2V}{3}mL\quad\frac{V}{3}mL$$

5.6 mL O_2 at STP/ 96500 C charge

$$\frac{V}{3}mL\ O_2\ at\ STP/\frac{96500}{5.6}\times\frac{V}{3}\ C\quad\text{charge}=5.74\times10^3\times V\ coulomb$$

10. D

fact

11. B

$$\sqrt{n\left(n+2\right)}=\sqrt{15}$$

$$n^2+2n-15=0$$

$$n=3$$

$$24^{Cr^{3+}}=1s^2,2s^2,2p^6,3s^2,3p^6,3d^3$$

Paired $e^-=18$

Unpaired $e^-=3$

12. C

$$\Delta H=705+1151-2\times328$$

$$=1200\ KJ$$

13. B
Fact

14. C

$1\,g\ x\%\ H_2O_2\ solution = \dfrac{x\times10^{-2}}{34}\ mole\ H_2O_2$

$H_2O_2 + KMnO_4 \xrightarrow{H^+} O_2 + M_n^{2+}$

$(n=2)\quad (n=5)$

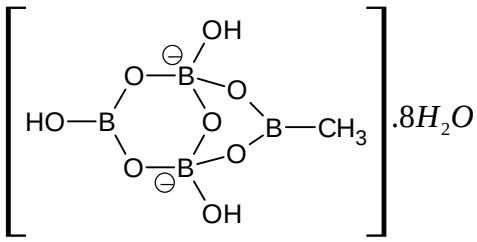
$\dfrac{x\times10^{-2}}{34}\ mole\ H_2O_2 \equiv \dfrac{2}{5}\times\dfrac{x\times10^{-2}}{34}\ mole\ KMnO_4$

$\dfrac{2}{5}\times\dfrac{x\times10^{-2}}{34} = molarity\times x\times10^{-3}$

Molarity = $\dfrac{2\times10}{5\times34}$

Normality $5\times\dfrac{2\times10}{5\times34} = \dfrac{20}{34}\ N = 0.58\ N \simeq 0.6N$

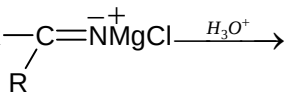
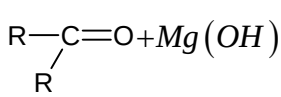
15. D

$2Na^+$  $.8H_2O$

Number of $B-O-B$ bonds $=5$
Number of $B-B$ bonds $=0$
Number of sp^2 B-atoms $=2$

16. A
A, due to aromaticity

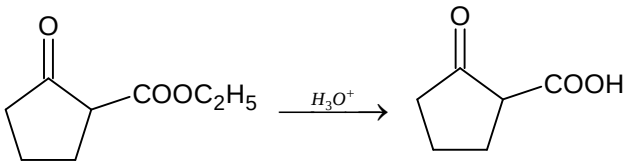
17. C

$RMgCl + RCN \longrightarrow$  $\xrightarrow{H_3O^+}$  $+ Mg(OH)Cl + NH_3$

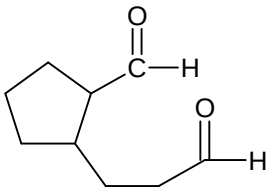
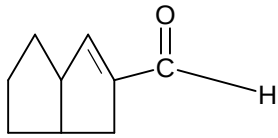
18. B

$R-C\equiv N \xrightarrow[(ii)\,H_2O/H^+]{(i)\,RMgX} R_2CO$

19. A



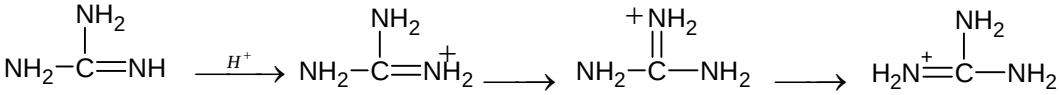
20. B

$X =$  ; $Y =$ 

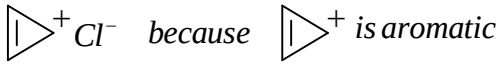
21. A
Cl^- replaces OH^- due to acidic medium (elimination reduction)

22. C
Tischenko's reaction

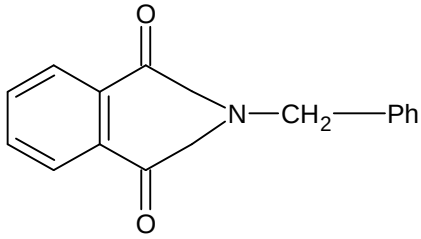
23. A
Protonation of base is stabilized by resonance.



24. B

 *because*

25. A

$A =$  ; $B = PhCH_2 - NH_2$

26. B
Fact

27. B
On the basis of resonance.

28. B
Fact

29. A
Fact

30. C
VSEPR theory

Mathematics

PART – III

SECTION - A

1. B

$4x^2 - 4x + 2 = \sin^2 y$

$\text{since } 0 \le \sin^2 y \le 1$

$\Rightarrow 0 < 4x^2 - 4x + 2 \le 1 \qquad (\because 4x^2 - 4x + 2 > 0)$

$\Rightarrow (2x-1)^2 \le 0$

$\Rightarrow x = \frac{1}{2}$

$\therefore \sin^2 y = 1$

$y = \pm\frac{\pi}{2}$

2. A

$\sin 47° - \sin 25° + \tan 61° - \sin 11°$

$= (\sin 61° + \sin 47°) - (\sin 25° + \sin 11°)$

$= 2\sin 54° \cos 7° - 2\sin 18° \cos 7°$

$= 2\cos 7°(\sin 54° - \sin 18°)$

$= 2\cos 7°(\cos 36° - \sin 18°)$

$= 2\cos 7°\left(\frac{\sqrt{5}+1}{4} - \frac{\sqrt{5}-1}{4}\right) = \cos 7°$

3. C

Let equation of tangent for $\left(ct, \frac{c}{t}\right)$ is $ty + \frac{x}{t} - 2c = 0$

Normal from P is $xt^3 - ty - ct^4 + c = 0$

Co – ordinate of T $(2ct, 0)$, $T'\left(0, \frac{2c}{t}\right)$, $N\left(\frac{ct^4 - c}{t^3}, 0\right)$ and $N'\left(0, \frac{c - ct^4}{t}\right)$

$\Rightarrow \Delta = \frac{c^2(1+t^4)}{2t^4}$ and $\left|\frac{1}{OA^2} - \frac{1}{OB^2}\right| = \frac{1}{a^2} - \frac{1}{b^2}$

$\frac{1}{\Delta} + \frac{1}{\Delta'} = \frac{2}{c^2}$

13. D

The point $([P+1], [P])$ lies inside the circle $x^2 + y^2 - 2x - 15 = 0$

$\therefore [P+1]^2 + [P]^2 - 2[P+1] - 15 < 0$

$\Rightarrow \quad [P]^2 < 8 \qquad$ ..... (1)

Point $([P+1].[P])$ lies outside the circle $x^2 + y^2 - 2x - 7 = 0$

$\Rightarrow [P+1]^2 + [P]^2 - 2[P+1] - 7 > 0$

$\Rightarrow [P]^2 > 4 \qquad$ ..... (2)

$4 < [P]^2 < 8$

14. B

Centres of circles are

$C_1(-\lambda,0), C_2(-\lambda_2,0), C_3(-\lambda_3,0)$

$\because OC_1, OC_2, OC_3$ are in GP

$\therefore \quad \lambda_2^2 = \lambda_1, \lambda_3$

Let any point on $x^2+y^2=c^2$ be $P(c\cos\alpha, c\sin\alpha)$,

$\therefore$ Length of tangents are

$PT_1 = \sqrt{2\lambda_1(c\cos\alpha)}$

$PT_2 = \sqrt{2\lambda_2(c\cos\alpha)}$

$PT_3 = \sqrt{2\lambda_3(c\cos\alpha)}$

$\Rightarrow PT_2^2 = PT_1, PT_3$

15. C

Equation of the radical axis is

$4\lambda x = 9 \quad or \quad x = \dfrac{9}{4\lambda}$

Pitting x in $x^2+y^2=1$

$y = \pm\sqrt{1-\dfrac{81}{16\lambda^2}}$

For the circle to have two common tangents, they must intersect at two distinct points

$\therefore \quad 1-\dfrac{81}{16\lambda^2} > 0$

$\Rightarrow \quad \lambda < \dfrac{-9}{4} \quad or \quad \lambda > \dfrac{9}{4}$

16. C

$y = |x-1|$

$\Rightarrow y^2 = (x-1)^2$

For image in y- axis, charge x by $-x$

$y^2 = (-x-1)^2$

$\Rightarrow x^2 - y^2 + 2x + 1 = 0$

17. D

Use $\Delta = 0$

$16\gamma^2 - 16p^2\gamma + 4p(\alpha p + \beta p - \alpha\beta) = 0$

Since γ is real

$\therefore D \geq 0$

$\Rightarrow (p-\alpha)(p-\beta) \geq 0$

18. A

$\because \dfrac{1}{10!}\left\{\dfrac{2\times 10!}{1!9!} + \dfrac{2\times 10!}{3!7!} + \dfrac{10!}{5!5!}\right\} = \dfrac{2^m}{n!}$

$\Rightarrow \dfrac{1}{10!}\left\{{}^{10}C_1 + {}^{10}C_3 + \ldots\ldots + {}^{10}C_9\right\} = \dfrac{2^m}{n!}$

$\Rightarrow \frac{1}{10!}2^9 = \frac{2^m}{n!}$

$\therefore m = 9 \;\; n = 10$

$\therefore$ Lines $x - y + 1 = 0 \;\&\; x + y + 3 = 0$ are perpendicular to each other

$\therefore$ Orthocentre $(-2, -1)$

$\therefore (2m - 2n, m - n)$

19. B

Since $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cos\frac{A}{2}$ $\left[\tan \textit{ gent rule}\right]$

$\therefore \tan\left(\frac{B-C}{2}\right)\cot\left(\frac{B+C}{2}\right) = \frac{b-c}{b+c}$

$\Rightarrow p = \frac{b-c}{b+c} \quad \therefore \frac{1+p}{1-p} = \frac{b}{c} - - - - - (1)$

Similarly $\frac{1+q}{1-q} = \frac{c}{a} - - - - (2)$

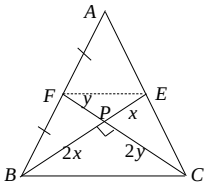
$\frac{1+p}{1-r} = \frac{a}{b} - - - (3)$

$\therefore \left(\frac{1+p}{1-p}\right)\left(\frac{1+q}{1-q}\right)\left(\frac{1+r}{1-r}\right) = 1$

$\Rightarrow 2(p+q+r) = -2pqr$

$\sum p = -pqr$

20. A



$\tan \angle PBF = \frac{y}{2x}$

$\tan \angle PBC = \frac{y}{x}$

$\tan B = \frac{\frac{y}{2x} + \frac{y}{x}}{1 - \frac{y^2}{2x^2}} = \frac{3xy}{2x^2 - y^2}$

$\cot B = \frac{2x^2 - y^2}{3xy}$

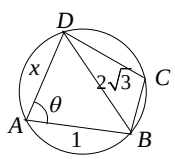
Similarly $\cot C = \frac{2y^2 - x^2}{3xy}$

$\cot B + \cot C = \frac{x^2 + y^2}{3xy} \geq \frac{2}{3}$

21. A

Let $AD = x$ and $\angle BAD = \theta$

In ΔABD, $\frac{BD}{\sin\theta} = 2R$

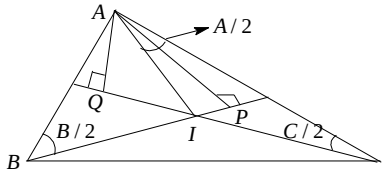


$\Rightarrow \sin\theta = \dfrac{2\sqrt{3}}{2\times 2} \;\therefore \theta = 60°$

$\cos 60° = \dfrac{x^2 + 1^2 - \left(2\sqrt{3}\right)^2}{2.x.1}$

$\Rightarrow x = \dfrac{1 \pm 3\sqrt{5}}{2} \quad \therefore x = \dfrac{1+3\sqrt{5}}{2} \;\left(\because x > 0\right)$

22. D



$\dfrac{AQ}{AC} = \sin\dfrac{C}{2} \;\Rightarrow AQ = AC\sin\dfrac{C}{2} .....(1)$

In ΔAIC, $\dfrac{CI}{\sin\dfrac{A}{2}} = \dfrac{AC}{\sin\left\{\pi - \left(\dfrac{A+C}{2}\right)\right\}}$

$\Rightarrow CI = AC\dfrac{\sin A/2}{\cos B/2} ----(2)$

$(1)/(2) \quad \dfrac{AQ}{CI} = \dfrac{\cos\dfrac{B}{2}\sin\dfrac{C}{2}}{\sin\dfrac{A}{2}}$

Similarly $\dfrac{AP}{BI} = \dfrac{\cos\dfrac{C}{2}\sin\dfrac{B}{2}}{\sin\dfrac{A}{2}}$

$\dfrac{AQ}{CI} + \dfrac{AP}{BI} = \dfrac{\sin\left(\dfrac{B}{2}+\dfrac{C}{2}\right)}{\sin\dfrac{A}{2}} = \tan\left(\dfrac{B+C}{2}\right)$

23. B

$\cos^2 x + \left(\dfrac{\sqrt{3}+1}{2}\right)\sin x - \dfrac{\sqrt{3}}{4} - 1 = 0$

$\Rightarrow 1 - \sin^2 x + \left(\dfrac{\sqrt{3}+1}{2}\right)\sin x - \dfrac{\sqrt{3}}{4} - 1 = 0$

$\Rightarrow \left(2\sin x - \sqrt{3}\right)\left(2\sin x - 1\right) = 0$

$\Rightarrow x = \dfrac{\pi}{3}, \dfrac{2\pi}{3}, \dfrac{\pi}{6}, \dfrac{5\pi}{6} \quad \because x \in [-\pi, \pi]$

24. A

$\tan\left(5\pi\cos\theta\right) = \tan\left\{\dfrac{\pi}{2} - 5\pi\sin\theta\right\}$

$\Rightarrow 5\pi\cos\theta = n\pi + \left(\dfrac{\pi}{2} - 5\pi\sin\theta\right)$

$\Rightarrow \cos\theta + \sin\theta = \dfrac{2n+1}{10}$

$\Rightarrow \sin\left(\theta+\frac{\pi}{4}\right)=\frac{2n+1}{10\sqrt{2}}$

$n=14,$ for each n thane are two values of θ

$\therefore 28\, solutions$

25. B

Let $\tan x=t$

$\therefore$ equation reduces to

$$\frac{4t^2}{1+t^2}+t^2+\frac{2}{t^2}=5$$

$\Rightarrow \left(t^2-1\right)\left(t^4+t^2-2\right)=0$

$\Rightarrow t^2=1,-2$

$x=\frac{\pi}{4},\frac{3\pi}{4}$

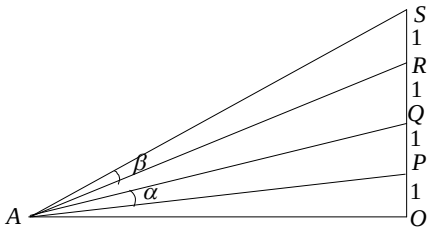
26. C

$\sin^2\left(\frac{x}{2}\right)=\frac{x}{(2n+1)\pi}$

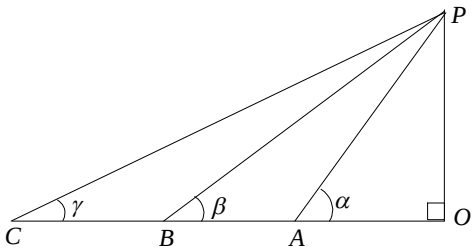
The last point after which graph of $y=\frac{x}{(2n+1)\pi}$ will stop cutting will be $(2n+1)\,\pi$

$\therefore$ Total intersection $=2(n+1)$

27. C



28. B



29. A

Putting n = 1 $P(1)=7^2+2^\circ.3^\circ=50=2\times 25$

Also confirm by $P(n)\Rightarrow P(n+1)$ induction step

30. D

$n(n+1)$ is always even. So given $P(n)$ is wrong.